

2 Confrontation of Classical Electrodynamics with Galilean Relativity

2.1 Galilean Transformations

Consider two systems of reference Σ and Σ' . Σ is characterized by coordinates (x, y, z) and time t , whereas Σ' is characterized by coordinates (x', y', z') and time t' . Let the origins of the two systems coincide at $t = t' = 0$ and Σ' is moving relative to Σ with constant velocity $\vec{v}$ in an arbitrary direction (see Fig. 1), although for simplicity we consider that the axes of the two frames are parallelly oriented.

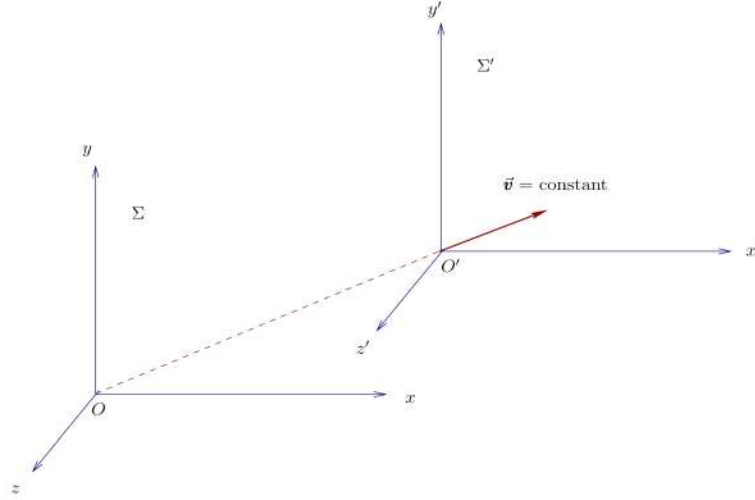


Figure 1: Galilean transformations for frame motion in an arbitrary direction, preserving the axes orientation.

The *Galilean transformation* equations represent a set of linear coordinate transformations given by

$$\begin{array}{|l} t' = t \\ x' = x - v_x t \\ y' = y - v_y t \\ z' = z - v_z t \end{array} \quad \text{or, in vector notation:} \quad \begin{array}{|l} t' = t \\ \vec{x}' = \vec{x} - \vec{v}t \end{array}, \quad (2.1)$$

where v_x, v_y, v_z are the components of $\vec{v}$ along x, y and z respectively.

2.2 Galilean Principle of Relativity

“All laws of mechanics are invariant under the Galilean transformations.”

2.2.1 Illustration: System of particles interacting via two-body central potentials

Equation of motion of the i^{th} particle in the reference frame Σ' :

$$m_i \frac{d\vec{v}'_i}{dt'} = - \nabla'_i \sum_j V_{ij} |\vec{x}'_i - \vec{x}'_j| \quad (2.2)$$

m_i being the mass of the i^{th} particle, which interacts with the j^{th} particle via the potential V_{ij} . From Eqs. (2.1) it follows:

$$t' = t, \quad \vec{x}'_i - \vec{x}'_j = \vec{x}_i - \vec{x}_j, \quad (2.3)$$

so that

$$\vec{v}'_i = \frac{d\vec{x}'_i}{dt'} = \frac{d\vec{x}_i}{dt} - \vec{v} = \vec{v}_i - \vec{v} \implies \frac{d\vec{v}'_i}{dt'} = \frac{d\vec{v}_i}{dt}. \quad (2.4)$$

Now, to work out the relations between the coordinate differentials in the two frames Σ and Σ' , let us consider a scalar function ψ [$\psi = \psi(t, x, y, z)$ in Σ , and $\psi = \psi(t', x', y', z')$ in Σ']. Then

$$d\psi = \frac{\partial\psi}{\partial t}dt + \frac{\partial\psi}{\partial x}dx + \frac{\partial\psi}{\partial y}dy + \frac{\partial\psi}{\partial z}dz = \frac{\partial\psi}{\partial t'}dt' + \frac{\partial\psi}{\partial x'}dx' + \frac{\partial\psi}{\partial y'}dy' + \frac{\partial\psi}{\partial z'}dz'. \quad (2.5)$$

Or,

$$\frac{\partial\psi}{\partial t}dt + \frac{\partial\psi}{\partial x}dx + \frac{\partial\psi}{\partial y}dy + \frac{\partial\psi}{\partial z}dz = \frac{\partial\psi}{\partial t'}dt + \frac{\partial\psi}{\partial x'}(dx - v_x dt) + \frac{\partial\psi}{\partial y'}(dy - v_y dt) + \frac{\partial\psi}{\partial z'}(dz - v_z dt). \quad (2.6)$$

Equating separately the coefficients of dt, dx, dy and dz on both sides, we have

$$\frac{\partial}{\partial t} = \frac{\partial}{\partial t'} - v_x \frac{\partial}{\partial x'} - v_y \frac{\partial}{\partial y'} - v_z \frac{\partial}{\partial z'}, \quad \frac{\partial}{\partial x} = \frac{\partial}{\partial x'}, \quad \frac{\partial}{\partial y} = \frac{\partial}{\partial y'}, \quad \frac{\partial}{\partial z} = \frac{\partial}{\partial z'}, \quad \text{and} \quad \frac{\partial}{\partial t} = \frac{\partial}{\partial t'} - v_x \frac{\partial}{\partial x'} - v_y \frac{\partial}{\partial y'} - v_z \frac{\partial}{\partial z'}. \quad (2.7)$$

Or, in vectorial form

$$\nabla = \nabla' ; \quad \frac{\partial}{\partial t} = \frac{\partial}{\partial t'} - \vec{v} \cdot \nabla', \quad (2.8)$$

and conversely,

$$\nabla' = \nabla , \quad \frac{\partial}{\partial t'} = \frac{\partial}{\partial t} + \vec{v} \cdot \nabla . \quad (2.9)$$

Using Eqs. (2.3), (2.4) and the first relation of (2.9), we can write Eq. (2.2) as

$$m_i \frac{d\vec{v}_i}{dt} = - \nabla_i \sum_j V_{ij} |\vec{x}_i - \vec{x}_j| \quad (2.10)$$

which is the equation of motion of the i^{th} particle in the reference frame Σ . This demonstrates that the Galilean transformation equations preserve the form of the Newtonian equations of motion.

2.2.2 Counter-illustration: Homogeneous wave equation for a scalar field

Let us again consider a scalar field ψ , and suppose that it satisfies the (homogeneous) wave equation in the reference frame Σ :

$$\left(\frac{1}{u^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \psi(t, \vec{x}) = 0, \quad (2.11)$$

where the speed of propagation of the wave is u .

Under the Galilean transformations, the use of the Eqs. (2.8) at once shows that the form of the above wave equation (2.11) is not preserved in the reference frame Σ' , if one imposes the condition that the propagation speed u should remain unchanged under such transformations:

$$\left[\frac{1}{u^2} \left\{ \frac{\partial^2}{\partial t'^2} - 2\vec{v} \cdot \nabla' \frac{\partial}{\partial t'} + (\vec{v} \cdot \nabla') (\vec{v} \cdot \nabla') \right\} - \nabla'^2 \right] \psi(t', \vec{x}') = 0. \quad (2.12)$$

2.3 The general incompatibility of Maxwell's Electromagnetism with Galilean Relativity

The above counter-illustration is an indication that the Galilean relativity is in general conflicting with the electrodynamic theory formulated by Maxwell. To see this, let us first show that the scalar ψ in (2.11) could be identified as any of the components of the electric field vector $\vec{E}$ or the magnetic field vector $\vec{B}$, which satisfy the Maxwell's equations in empty space:

$$\left. \begin{array}{ll} \text{(i)} \quad \nabla \cdot \vec{E} = 0 & \text{(iii)} \quad \nabla \times \vec{E} + \frac{1}{c} \frac{\partial \vec{B}}{\partial t} = 0 \\ \text{(ii)} \quad \nabla \cdot \vec{B} = 0 & \text{(iv)} \quad \nabla \times \vec{B} - \frac{1}{c} \frac{\partial \vec{E}}{\partial t} = 0 \end{array} \right\}. \quad (2.13)$$

Taking the curl of both sides of Eq. (2.13 iii), one gets

$$\nabla \times (\nabla \times \vec{E}) = -\frac{1}{c} \nabla \times \frac{\partial \vec{B}}{\partial t} \implies \nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\frac{1}{c} \frac{\partial}{\partial t} \nabla \times \vec{B}. \quad (2.14)$$

Then the use of Eqs. (2.13 i) and (2.13 iv) leads to

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \vec{E} = 0. \quad (2.15)$$

Similarly, taking the curl of Eq. (2.13 iv) and using Eqs. (2.13 ii) and (2.13 iii) we get

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \vec{B} = 0. \quad (2.16)$$

Each of the component equations of (2.15) and (2.16) are of the form of Eq. (2.11)] with $u \equiv c$:

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \psi(t, \vec{x}) = 0, \quad [\psi := \text{any component of } \vec{E} \text{ or } \vec{B}]. \quad (2.17)$$

This is regarded as the homogeneous wave equation for the following reason:

Assume a solution of Eq. (2.17) of the form

$$\psi(t, \vec{x}) = \psi_0 e^{i(\vec{k} \cdot \vec{x} - \omega t)}, \quad (2.18)$$

where $\vec{k}$ is a constant vector, and ω and ψ_0 are also constants. We have

$$\nabla \psi = i\vec{k}\psi \implies \nabla^2 \psi = -|\vec{k}|^2 \psi \quad \text{and} \quad \frac{\partial \psi}{\partial t} = -i\omega \psi \implies \frac{\partial^2 \psi}{\partial t^2} = -\omega^2 \psi, \quad (2.19)$$

which implies that

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) \psi = \left(-\frac{\omega^2}{c^2} + |\vec{k}|^2 \right) \psi = 0, \quad (2.20)$$

under the stipulation

$$k^2 \equiv |\vec{k}|^2 = \frac{\omega^2}{c^2}. \quad (2.21)$$

The solution (2.18) represents a wave “traveling” along (or opposite to) $\vec{k}$ (known as the *wave vector*) in the sense that a point of constant phase $(\vec{k} \cdot \vec{x} - \omega t)$ moves in the direction of $\pm \vec{k}$ with a speed c , which is determined by

$$\vec{k} \cdot \vec{x} - \omega t = \text{constant} \implies \vec{k} \cdot \frac{d\vec{x}}{dt} = \omega \quad \text{or, by Eq. (2.21):} \quad c = \pm \frac{\omega}{k} = \pm \hat{k} \cdot \frac{d\vec{x}}{dt}, \quad (2.22)$$

where $\hat{k} = \vec{k}/|\vec{k}|$ is the unit vector along $\vec{k}$. Moreover, since a surface of constant phase, given by the equation $\vec{k} \cdot \vec{x} - \omega t = \text{constant}$, is a *plane* perpendicular to $\vec{k}$, the solution (2.18) represents a *plane wave*.

Now, the non-covariance of the wave equation under the Galilean transformations (as shown above) poses a problem for Maxwell's electromagnetism, unlike for instance the theory of acoustics. The acoustic (sound) waves are due to the compressions and rarefactions in a given medium (e.g. air), and the reference frame Σ in which the wave equation has its usual form [Eq. (2.11)] is obviously a preferred one in which the medium is at rest. In all other reference frames, in which the medium is moving, the direct influence of such motion on the propagation of sound (by virtue of mechanical vibrations described by Newtonian classical mechanics) can account for the extra terms in Eq. (2.12) that modify the usual form of the wave equation under the Galilean transformations. For the electromagnetic waves however, the situation is in general different, because there is no influence of any intervening medium on the electromagnetic wave propagation other than just supporting the latter. The Maxwell's equations, which govern the electromagnetic wave propagation, involve the parameter c , which according to Maxwell's theory should remain “fixed” for all frames of reference, and could be interpreted as the fixed speed of propagation of the electromagnetic waves. It is the presence of this fixed c which prevents any possible way to compensate for the non-preservation of the Maxwell's equations (and consequently the wave equation) under the Galilean transformations. Maxwell's theory thus seems to be *incompatible* with Galilean relativity.

2.4 Ways to look for a consistent Relativistic theory of Electromagnetism

The following are the possible alternatives that one may choose to resolve the conflict between Maxwell's electromagnetism and Galilean relativity:

- I. Maxwell's theory is incorrect and a proper theory of electromagnetism must have its equations invariant under the Galilean transformations.
- II. Galilean relativity applies to the laws of Newtonian mechanics, and Maxwell's theory is correct but bizarre, in the sense that the electromagnetic wave propagation always requires a preferred frame — the one in which the intervening medium, the so-called luminiferous “*Æther*”, is at rest.
- III. There should exist a relativity principle, other than the Galilean one, for the laws of mechanics as well as the equations of electrodynamics.

The first alternative is hardly viable, given the immense success of the Maxwell's theory otherwise.

The no-go establishment of the *Æther*-hypothesis, following the works of Michaelson and Morley in 1886, rules out the second alternative as well.

So, what remained is to find out a set of transformation relations and a new relativity principle, so that the Maxwell's equations are of the same form, with the same speed parameter c in all frames moving with uniform relative speeds (i.e. inertial frames).

The first step towards obtaining such transformation relations, and hence to formulate a new theory of relativity, is to abolish the Newtonian concept of “*absolute space*”, and to keep the time and the space in equal footing. That is to say, one should now invoke the concept of “*space-time*” (instead of treating space and time separately) represented by all the spatial coordinates, as well as time multiplied by a constant speed (from dimensional arguments). Such a speed parameter must be a universal constant, which is argued to be the speed c of electromagnetic signals in empty space, as we see below.

3 Formulation of Special Relativity

3.1 The Postulates

I. “All laws of physics (except that for gravitation) are independent of uniform translational motion of the system, i.e., the reference frame (RF), in which they operate” — *Principle of Relativity*:

$$\text{Physical Laws} \xLeftrightarrow{\text{Inv.}} \left[\Sigma_{(RF)} \xrightarrow{v=\text{const.}} \Sigma'_{(RF)} \right]. \quad (3.1)$$

Corollary: The mathematical equations expressing the laws of nature must be *covariant*, i.e., invariant in form under the transformations that leave invariant the infinitesimal separation between two events in space-time.

II. “The speed of propagation (c) of electromagnetic signals in empty space is finite and independent of the motion of the source”:

$$c \xLeftrightarrow{\text{Inv.}} \left[\Sigma_{(RF)} \xrightarrow{v=\text{const.}} \Sigma'_{(RF)} \right]. \quad (3.2)$$

Corollary: For physical entities in every inertial frame, there is a finite limiting speed, equal to c (as argued via certain thought experiments):

$$v_{lim} = c = \text{const.} \quad (3.3)$$

The most general set of coordinate transformations which preserve the infinitesimal separation between two space-time events, side-by-side maintaining the constancy of c , are the *Poincaré transformations*. A sub-class of these are the so-called (*homogeneous*) *Lorentz transformations* which preserve the separation even when it is not necessarily infinitesimal, i.e., the events are not necessarily close enough. We discuss below in detail the form and properties of both the Lorentz and the Poincaré transformation equations in certain simplified, as well as generic, scenarios.

3.2 Lorentz Transformations for a Boost along one axis

3.2.1 Derivation of the Lorentz Transformation Equations

Consider two reference frames Σ and Σ' represented by (ct, x, y, z) and (ct', x', y', z') respectively, as shown in Fig. 2. Let us suppose that

- The axes of the two frames are parallelly oriented, i.e, $ct' || ct, x' || x, y' || y, z' || z$.
- Frame Σ' moves with constant velocity $\vec{v}$ relative to the frame Σ along, say, the positive x axis.
- The origins of the two frames coincide at $t = t' = 0$.

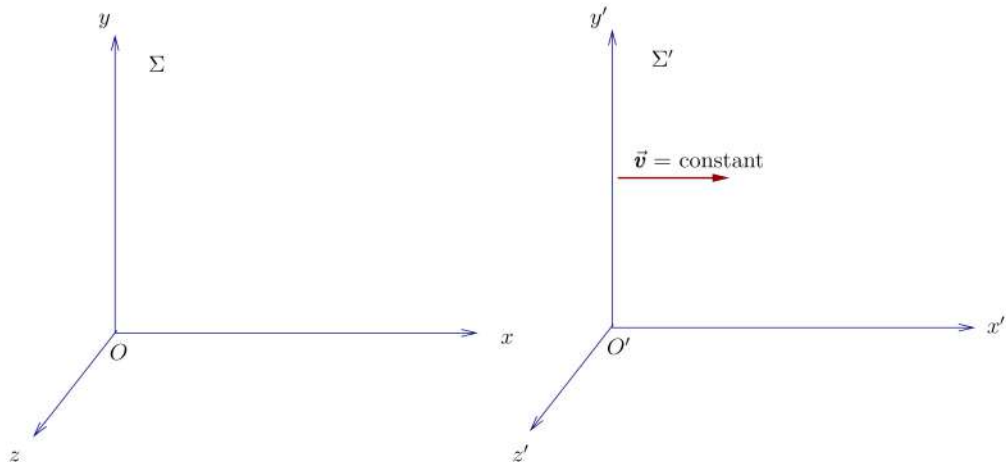


Figure 2: Lorentz transformations for frame motion along one axis, preserving the axes orientation.

Consider now an event P, whose coordinates are (ct', x', y', z') in frame Σ' and (ct, x, y, z) in frame Σ . As the origin of Σ' moves relative to Σ with speed v along positive x -direction

$$x' = 0 \quad \implies \quad x = vt. \quad (3.4)$$

Conversely, as the origin of Σ moves relative to Σ' with speed v along the negative x' -direction

$$x = 0 \quad \implies \quad x' = -vt. \quad (3.5)$$

Now, the transformation equations between the two systems must be *linear*, so that for an event in Σ there is a unique associated event in Σ' , and vice versa. The most general linear relations satisfying Eqs. (3.4) and (3.5) are

$$x' = \gamma(x - vt) \quad , \quad x = \gamma(x' + vt') \quad [\gamma = \text{dimensionless constant}]. \quad (3.6)$$

Or,

$$dx' = \gamma(dx - vdt) \quad , \quad dx = \gamma(dx' + vdt'). \quad (3.7)$$

Therefore,

$$\frac{dx'}{dt'} = \gamma \left(\frac{dx}{dt} - v \frac{dt}{dt'} \right) = \gamma \left(\frac{dx}{dt} - v \right) \frac{dt}{dt'} \quad , \quad \frac{dx}{dt} = \gamma \left(\frac{dx'}{dt'} + v \frac{dt'}{dt} \right) = \gamma \left(\frac{dx'}{dt'} + v \right) \frac{dt'}{dt}. \quad (3.8)$$

In the limit $dx'/dt' \rightarrow c$ and $dx/dt \rightarrow c$, Eqs. (3.8) reduce to

$$\frac{dt}{dt'} = \frac{c}{\gamma(c - v)} \quad , \quad \frac{dt'}{dt} = \frac{c}{\gamma(c + v)}. \quad (3.9)$$

Solving for γ , subject to the requirement that the transformation equations we seek must reduce to the Galilean transformation equations in the limit $v/c \rightarrow 0$, we get

$$\boxed{\gamma = (1 - \beta^2)^{-1/2}} \quad , \quad \text{where} \quad \boxed{\beta = \frac{v}{c}}. \quad (3.10)$$

One can now write down the full set of transformation equations by solving Eqs.(3.6) for x' and t' :

$$\boxed{\begin{aligned} \text{(i)} \quad ct' &= \gamma(ct - \beta x) \\ \text{(ii)} \quad x' &= \gamma(-\beta ct + x) \\ \text{(iii)} \quad y' &= y \\ \text{(iv)} \quad z' &= z \end{aligned}} \quad (3.11)$$

These are the *Lorentz transformation* (LT) equations in a simple scenario where there is a frame motion, or *Lorentz boost*, along one spatial direction (x here, note that there is no frame motion along y or z).

3.2.2 The Lorentz Transformation Matrix

Let us re-write the above LT equations (3.11) in matrix form as

$$\begin{pmatrix} x'^0 \\ x'^1 \\ x'^2 \\ x'^3 \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}. \quad (3.12)$$

where we denote the coordinates as:

$$x^0 \equiv ct \quad , \quad x^1 \equiv x \quad , \quad x^2 \equiv y \quad , \quad \text{and} \quad x^3 \equiv z \quad (3.13)$$

In compact form the LT equations are given by

$$\boxed{x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu}} \quad , \quad [\mu, \nu = 0, 1, 2, 3] , \quad (3.14)$$

where x^{μ} and Λ^{μ}_{ν} are the elements of

$$\underline{\mathbf{x}} \equiv \{x^{\mu}\} = \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix} : \quad 1 \times 4 \text{ Position (column) vector in } (3+1)\text{-dimensional spacetime} , \quad (3.15)$$

$$\underline{\mathbf{\Lambda}} \equiv \{\Lambda^{\mu}_{\nu}\} = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} : \quad \begin{array}{l} 4 \times 4 \text{ (homogeneous) LT matrix in a } (3+1)\text{-dimensional} \\ \text{space-time for single Lorentz boost along positive } x^1\text{-axis;} \\ \mu \text{ denotes the columns and } \nu \text{ denotes the rows.} \end{array} \quad (3.16)$$

In the Eq. (3.14), *Einstein's summation convention* is implied, i.e., repeated indices are summed over:

$$\Lambda^{\mu}_{\nu} x^{\nu} \equiv \sum_{\nu=0}^3 \Lambda^{\mu}_{\nu} x^{\nu} \quad [\forall \mu = 0, 1, 2, 3] . \quad (3.17)$$

The inverse transformation equations are given by

$$\boxed{x_{\mu} = \Lambda_{\mu}^{\nu} x'_{\nu}} \quad , \quad [\mu, \nu = 0, 1, 2, 3] , \quad (3.18)$$

where x^{μ} and Λ^{μ}_{ν} are the elements of

$$\underline{\mathbf{x}}^{-1} \equiv \{x_{\mu}\} = (x_0, x_1, x_2, x_3) \quad : \quad 4 \times 1 \text{ Position (row) vector in } (3+1)\text{-dimensional spacetime} , \quad (3.19)$$

$$\underline{\mathbf{\Lambda}}^{-1} \equiv \{\Lambda_{\mu}^{\nu}\} = \{[\Lambda^{\mu}_{\nu}]^{-1}\} \quad : \quad 4 \times 4 \text{ Inverse LT matrix for a boost along negative } x^1 \text{ axis} . \quad (3.20)$$

The distinction between x^0 and x_0 , x^1 and x_1 , and so forth, will be explained later on, when we will make a general study of tensors and their components. For the time being, however, let's stick to the matrix terminology and focus on the properties of the LT matrix Λ^{μ}_{ν} , where μ denotes columns and ν denotes rows. Following are the two important properties of the LT matrix for a single Lorentz boost:

$$(i) \quad \boxed{\underline{\mathbf{\Lambda}}^T = \underline{\mathbf{\Lambda}}} \implies \boxed{\underline{\mathbf{\Lambda}}^T \underline{\mathbf{\Lambda}}^{-1} = \underline{\mathbf{I}}} \implies \boxed{\Lambda^{\mu}_{\nu} \Lambda_{\mu}^{\alpha} = \delta_{\nu}^{\alpha}} , \quad \text{and} \quad (ii) \quad \boxed{\det \underline{\mathbf{\Lambda}} = 1} . \quad (3.21)$$

3.2.3 Lorentz Transformation as a Rotation: The Rapidity parameter

To observe the geometrical meaning of a “boost”, let us make the following parametrization:

$$\boxed{\beta = \frac{v}{c} = \tanh \zeta \quad , \quad \gamma = (1 - \beta^2)^{-1/2} = \cosh \zeta} , \quad (3.22)$$

where ζ is a dimensionless constant, called the ‘Boost’ or ‘Rapidity’ parameter. Under the above parametrization, the LT equations (3.11) take the form:

$$\begin{aligned} x'^0 &= x^0 \cosh \zeta - x^1 \sinh \zeta , \\ x'^1 &= -x^0 \sinh \zeta + x^1 \cosh \zeta , \\ x'^2 &= x^2 , \quad \text{and} \quad x'^3 = x^3 , \end{aligned} \quad (3.23)$$

i.e., the LT matrix (due to a *boost* along the positive x^1 -direction with constant speed v) takes the form:

$$\underline{\mathbf{\Lambda}} \equiv \{\Lambda^{\mu}_{\nu}\} = \begin{pmatrix} \cosh \zeta & -\sinh \zeta & 0 & 0 \\ -\sinh \zeta & \cosh \zeta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \quad (3.24)$$

Under a further transformation, and reparameterization:

$$x^0 \rightarrow \tilde{x}^0 = i x^0 \text{ (i.e., } t \rightarrow \tilde{t} = i t) ; \quad x^k \rightarrow \tilde{x}^k = x^k \quad [k = 1, 2, 3] , \quad \zeta = i \theta , \quad (3.25)$$

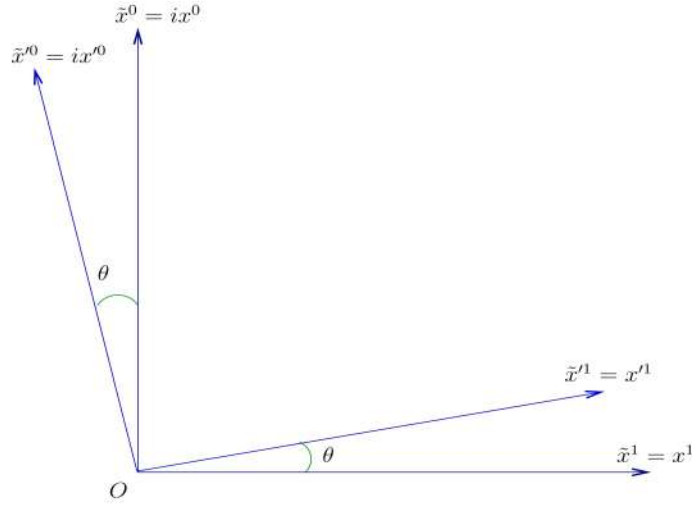


Figure 3: Simple Lorentz transformation as an (orthonormal) rotation of the $ix^0 - x^1$ plane.

the above LT matrix (3.24) reduces to

$$\underline{\Lambda} \equiv \{\Lambda^\mu_\nu\} = \begin{pmatrix} \cos \theta & \sin \theta & 0 & 0 \\ -\sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \{\mathcal{R}^\mu_\nu\} , \quad (3.26)$$

where $\mathcal{R}^\mu_\nu$ are the elements of the matrix corresponding to the ordinary (orthonormal) rotation of the $\tilde{x}^0 - \tilde{x}^1$ plane in the $(3+1)$ -dimensional space-time (see Fig. 3).

A *Lorentz Boost* along, say the positive x^1 direction, with speed v , may thus be viewed as a rotation of the complex $ix^0 - x^1$ plane by a real angle θ . In principle, there could be *three* Lorentz boosts along each of the three spatial axes. The transformations (3.25) imply the *Euclidianization* of space-time.

3.3 The invariant Line element

Refer back to the LT equations (3.11) for a boost along the positive x -direction with constant speed v . In differential form, they are given by

$$\begin{array}{ll} \text{(i)} & cdt' = \gamma(cdt - \beta dx) \\ \text{(ii)} & dx' = \gamma(-\beta cdt + dx) \\ \text{(iii)} & dy' = dy \\ \text{(iv)} & dz' = dz \end{array} \quad (3.27)$$

Now,

$$\begin{array}{lll} \text{Eq. (3.27 i)} + \text{Eq. (3.27 ii)} & \implies & cdt' + dx' = \gamma(1 - \beta)(cdt + dx) , \\ \text{Eq. (3.27 i)} - \text{Eq. (3.27 ii)} & \implies & cdt' - dx' = \gamma(1 + \beta)(cdt - dx) . \end{array}$$

Multiplying these two equations, and using the expressions (3.10) for β and γ , we get

$$c^2 dt'^2 - dx'^2 = c^2 dt^2 - dx^2 .$$

Using further, the Eqs. (3.27 iii & iv):

$$c^2 dt'^2 - dx'^2 - dy'^2 - dz'^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2 , \quad (3.28)$$

i.e., the Lorentz transformations leave invariant the differential quantity

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2 \equiv (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2. \quad (3.29)$$

Performing a similar operation as above on the set of LT equations (3.11) themselves, we could have also found that the Lorentz transformations leave invariant the algebraic quantity

$$s^2 = c^2 t^2 - x^2 - y^2 - z^2 \equiv (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2. \quad (3.30)$$

As in the ordinary Euclidean (three-)space, the elementary distance between two points is given by

$$|d\vec{x}| = \sqrt{dx^2 + dy^2 + dz^2} \equiv \sqrt{(dx^1)^2 + (dx^2)^2 + (dx^3)^2}, \quad (3.31)$$

the quantity ds^2 given by Eq. (3.29) is analogically referred to as the *Line element*, or the (squared) elementary separation between two events in a $(3+1)$ -dimensional space-time. Galilean transformations leave $|d\vec{x}|^2$ invariant, whereas Lorentz transformations leave ds^2 invariant.

Naïvely, the relative minus sign between the temporal and spatial parts of $ds^2 = c^2 dt^2 - |d\vec{x}|^2$ may be looked upon as follows: For signals travelling with the invariant speed c , which may also be identified with the speed of light in free space, Galilean relativity alludes to instantaneous reception after transmission, i.e., $\Delta t = 0$, which implies

$$\lim_{\Delta t \rightarrow 0} \frac{|\Delta \vec{x}|}{\Delta t} = \frac{|d\vec{x}|}{dt} = c \rightarrow \infty. \quad (3.32)$$

However, for finite c (as demanded in Maxwell's theory), the reception and the transmission should be separated in time by $\Delta t \neq 0$. One therefore imposes (instead of $dt = 0$)

$$ds^2 = c^2 dt^2 - |d\vec{x}|^2 = 0, \quad \text{so that} \quad \frac{|d\vec{x}|}{dt} = c, \quad \text{automatically.}$$

3.4 Poincaré Transformations for a Boost along one axis

The necessary condition for the constancy of c , in a relativity theory governed by certain transformation equations, is that the line element ds^2 must remain invariant under such transformations. Lorentz transformations (LT) preserve not only ds^2 , but s^2 [Eq. (3.30)] as well, which is not necessarily required. The invariance of s^2 is due to the third assumption made in the derivation of the LT equations in subsection 3.2, viz., the frames of reference Σ and Σ' coincide at $t = t' = 0$. If we relax this assumption, then the most general linear transformations that leave ds^2 invariant are the Poincaré transformations (PT), which in the simple case of a single Lorentz boost along (say) the x -axis, with constant speed v , are given by the equations:

$$\begin{aligned} \text{(i)} \quad ct' &= \gamma(ct - \beta x) + a_t \\ \text{(ii)} \quad x' &= \gamma(-\beta ct + x) + a_x \\ \text{(iii)} \quad y' &= y + a_y \\ \text{(iv)} \quad z' &= z + a_z \end{aligned}$$

(3.33)

where (a_t, a_x, a_y, a_z) are constant shifts between the origins of Σ and Σ' at $t = 0$, along the t, x, y and z directions respectively. The above equations are expressed in compact notation as

$$x'^\mu = \Lambda^\mu_\nu x^\nu + a^\mu, \quad (3.34)$$

where the elements of a^μ are

$$a^0 \equiv a_t, \quad a^1 \equiv a_x, \quad a^2 \equiv a_y, \quad a^3 \equiv a_z. \quad (3.35)$$

In differential form, however, the Lorentz and the Poincaré transformation equations are identical, viz.

$$dx'^\mu = \Lambda^\mu_\nu dx^\nu. \quad (3.36)$$

3.5 Lorentz Transformations for a Boost in an arbitrary direction

Consider again two frames Σ and Σ' with coordinates (x^0, x^1, x^2, x^3) and (x'^0, x'^1, x'^2, x'^3) respectively.

- The axes of Σ and Σ' are parallelly oriented, i.e. $x'^0 \parallel x^0$, $x'^1 \parallel x^1$, $x'^2 \parallel x^2$, $x'^3 \parallel x^3$.
- Frame Σ' moves with constant velocity $\vec{v}$ relative to the frame Σ in an arbitrary direction.
- The origins of Σ and Σ' coincide at $t = t' = 0$.

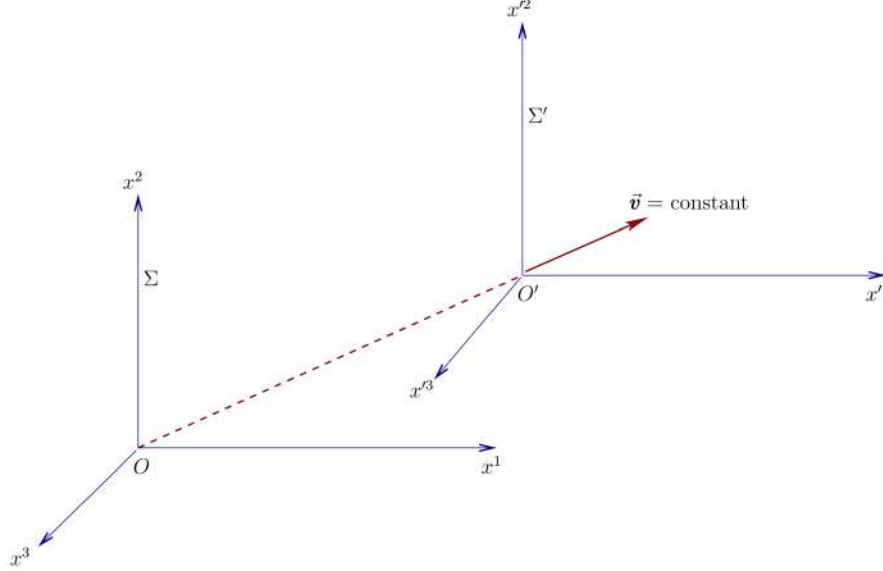


Figure 4: Lorentz transformation for frame motion in an arbitrary direction, preserving the axes orientation.

The corresponding Lorentz transformation equations involve three boost parameters, viz. the components (v^1, v^2, v^3) of the velocity $\vec{v}$ along respectively the three axes (x^1, x^2, x^3) . The elements of the general Lorentz boost transformation matrix are found in the Appendix A (see Eqs. (A.6) and (A.8)), one may verify that the corresponding transformation equations can be expressed in the ordinary (three-)vectorial notation as

$$\boxed{x'^0 = \gamma (x^0 - \vec{\beta} \cdot \vec{x}), \quad \vec{x}' = \vec{x} + \frac{(\gamma - 1)}{|\vec{\beta}|^2} (\vec{\beta} \cdot \vec{x}) \vec{\beta} - \vec{\beta} \gamma x^0}, \quad (3.37)$$

where

$$\vec{\beta} = (\beta^1, \beta^2, \beta^3) = \frac{\vec{v}}{c} = \left(\frac{v^1}{c}, \frac{v^2}{c}, \frac{v^3}{c} \right), \quad \gamma = (1 - |\vec{\beta}|^2)^{-1/2}. \quad (3.38)$$

3.6 Lorentz Transformations for arbitrary Boosts and Rotations

If, in addition to the scenario in previous subsection, the axes of the two frames Σ and Σ' are not taken to be parallelly oriented (see Fig. 5), then the corresponding LT matrix is in its most general form, which involves not only the three boost parameters (speeds), but three spatial rotation parameters (angles) as well. Even for this most general LT matrix, the first of the properties (3.21), viz.,

$$\boxed{\underline{\Lambda}^T \underline{\Lambda}^{-1} = \underline{\mathbf{I}}} \quad \Longleftrightarrow \quad \boxed{\Lambda^\mu{}_\nu \Lambda^\alpha{}_\mu = \delta_\nu^\alpha}, \quad (3.39)$$

can be shown to hold. As to the second property, taking determinant of both sides of Eq.(3.39), we have

$$\boxed{\det \underline{\Lambda} = \pm 1}. \quad (3.40)$$

Our interest is however in the so-called *proper* Lorentz transformations for which

$$\Lambda^0_0 \geq 0, \quad \text{and} \quad \det \underline{\Lambda} = +1, \quad (3.41)$$

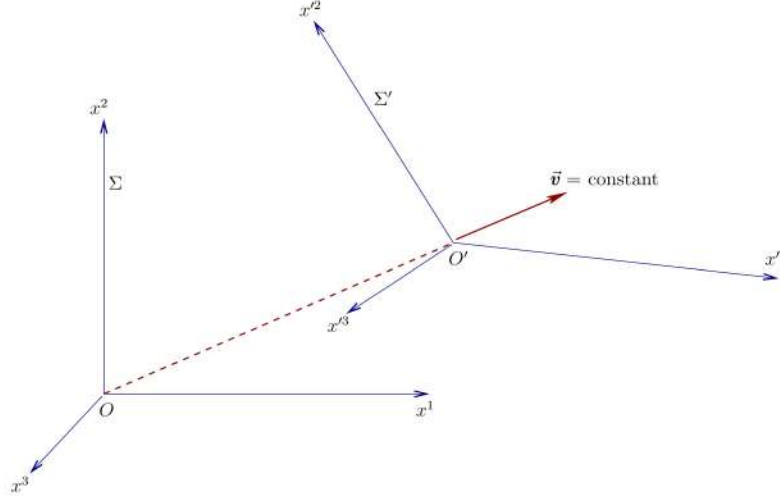


Figure 5: Lorentz transformation for frame motion in an arbitrary direction, and not preserving the axes orientation.

and which can always be converted to the identity transformation by continuous variation of parameters of Λ^μ_ν . Actually one can decompose the full LT into an ordinary spatial rotation, followed by a boost, followed by a further ordinary rotation. The first rotation lines up one of the spatial axes (say the x^1 axis) of Σ with the velocity $\vec{v}$ of Σ' . Then a boost in this direction with speed $v = |\vec{v}|$ transforms Σ to a frame which is at rest relative to Σ' . Finally, another rotation lines up the coordinate frame of Σ with that of Σ' .

3.7 Successive Lorentz Boosts

Two frames Σ and Σ' , with coordinates (x^0, x^1, x^2, x^3) and (x'^0, x'^1, x'^2, x'^3) , have their axes parallelly oriented and origins coincident at $x^0 = x'^0 = 0$. Frame Σ' moves with constant velocity $\vec{v}$ in an arbitrary direction, relative to the frame Σ .

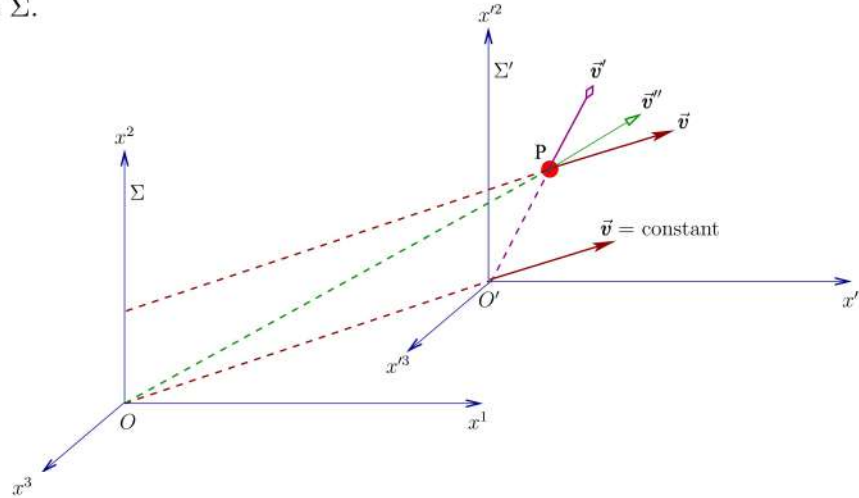


Figure 6: Successive Lorentz Boosts

Consider now a particle P which moves with constant velocity $\vec{v}'$ relative to Σ' . Let $\vec{v}''$ be the particle's velocity relative to Σ , and Σ'' be the particle's rest frame, i.e, the coordinate system $(x''^0, x''^1, x''^2, x''^3)$ attached to the particle. Then

I. For the relative motion between Σ' and Σ :

$$x'^\mu = \Lambda^\mu_\nu(\vec{\beta}) x^\nu, \quad [\vec{\beta} = \vec{v}/c]. \quad (3.42)$$

where $\Lambda^\mu_\nu(\vec{\beta})$ are the corresponding LT matrix elements.

II. For the relative motion between Σ'' and Σ' :

$$x''^\mu = \Lambda^\mu_\nu(\vec{\beta}') x'^\nu, \quad [\vec{\beta}' = \vec{v}'/c]. \quad (3.43)$$

where $\Lambda^\mu_\nu(\vec{\beta}')$ are the corresponding LT matrix elements.

III. For the relative motion between Σ'' and Σ :

$$x''^\mu = \Lambda^\mu_\nu(\vec{\beta}'') x^\nu, \quad [\vec{\beta}'' = \vec{v}''/c]. \quad (3.44)$$

where $\Lambda^\mu_\nu(\vec{\beta}'')$ are the corresponding LT matrix elements.

Substituting for x'^ν from Eq. (3.42) in Eq. (3.43), and comparing with Eq. (3.44), we have

$$x''^\mu = \Lambda^\mu_\lambda(\vec{\beta}') \Lambda^\lambda_\nu(\vec{\beta}) x^\nu = \Lambda^\mu_\nu(\vec{\beta}'') x^\nu \implies \boxed{\Lambda^\mu_\nu(\vec{\beta}'') = \Lambda^\mu_\lambda(\vec{\beta}') \Lambda^\lambda_\nu(\vec{\beta})}. \quad (3.45)$$

Thus a Lorentz boost along the direction of $\vec{\beta}''$ is the product of the Lorentz boosts along $\vec{\beta}'$ and $\vec{\beta}$.

3.7.1 Non-commutativity of the products of Boosts

The product of the boosts is in general non-commutative:

$$\Lambda^\mu_\lambda(\vec{\beta}') \Lambda^\lambda_\nu(\vec{\beta}) \neq \Lambda^\mu_\lambda(\vec{\beta}) \Lambda^\lambda_\nu(\vec{\beta}'). \quad (3.46)$$

To verify this, let us introduce a boost parameter $\vec{\beta}''$ such that

$$\Lambda^\mu_\nu(\vec{\beta}'') = \Lambda^\mu_\lambda(\vec{\beta}) \Lambda^\lambda_\nu(\vec{\beta}'). \quad (3.47)$$

We need to check under what circumstances $\Lambda^\mu_\nu(\vec{\beta}'') \neq \Lambda^\mu_\nu(\vec{\beta}'')$. Let us consider the following simple cases:

Case I: $\vec{\beta} \parallel \vec{\beta}'$

Let us choose $\vec{\beta}$ and $\vec{\beta}'$ to be directed along the positive x^1 axis, i.e.

$$\vec{\beta} = (\beta, 0, 0) \quad \text{and} \quad \vec{\beta}' = (\beta', 0, 0), \quad (3.48)$$

whence the non-vanishing elements of $\Lambda^\mu_\nu(\vec{\beta})$ and $\Lambda^\mu_\nu(\vec{\beta}')$ are given by

$$\Lambda^\mu_\nu(\vec{\beta}) : \quad \begin{cases} \Lambda^0_0(\beta) = \Lambda^1_1(\beta) = \gamma, \\ \Lambda^1_0(\beta) = \Lambda^0_1(\beta) = -\beta\gamma, \\ \Lambda^2_2(\beta) = \Lambda^3_3(\beta) = 1, \end{cases} \quad \left[\gamma = \frac{1}{\sqrt{1-\beta^2}} \right], \quad (3.49)$$

$$\Lambda^\mu_\nu(\vec{\beta}') : \quad \begin{cases} \Lambda^0_0(\beta') = \Lambda^1_1(\beta') = \gamma', \\ \Lambda^1_0(\beta') = \Lambda^0_1(\beta') = -\beta'\gamma', \\ \Lambda^2_2(\beta') = \Lambda^3_3(\beta') = 1, \end{cases} \quad \left[\gamma' = \frac{1}{\sqrt{1-\beta'^2}} \right]. \quad (3.50)$$

Using Eqs. (3.45) and (3.47) the non-vanishing elements of $\Lambda^\mu_\nu(\vec{\beta}'')$ and $\Lambda^\mu_\nu(\vec{\beta}'')$ are computed as follows:

$$\text{0-0 element: } \begin{cases} \Lambda^0_0(\vec{\beta}'') = \Lambda^0_\lambda(\vec{\beta}') \Lambda^\lambda_0(\vec{\beta}) \\ = \Lambda^0_0(\beta') \Lambda^0_0(\beta) + \Lambda^1_1(\beta') \Lambda^1_0(\beta) + 0 + 0 \\ = \gamma'\gamma(1 + \beta'\beta) \\ \Lambda^0_0(\vec{\beta}'') = \Lambda^0_\lambda(\vec{\beta}) \Lambda^\lambda_0(\vec{\beta}') \\ = \Lambda^0_0(\beta) \Lambda^0_0(\beta') + \Lambda^1_1(\beta) \Lambda^1_0(\beta') + 0 + 0 \\ = \gamma\gamma'(1 + \beta\beta') \end{cases} \implies \Lambda^0_0(\vec{\beta}'') = \Lambda^0_0(\vec{\beta}''). \quad (3.51)$$

$$\begin{aligned}
\text{1-0 element: } \left\{ \begin{array}{l} \Lambda^1_0(\vec{\beta}'') = \Lambda^1_\lambda(\vec{\beta}') \Lambda^\lambda_0(\vec{\beta}) \\ \phantom{\Lambda^1_0(\vec{\beta}'')} = \Lambda^1_0(\beta') \Lambda^0_0(\beta) + \Lambda^1_1(\beta') \Lambda^1_0(\beta) + 0 + 0 \\ \phantom{\Lambda^1_0(\vec{\beta}'')} = -\gamma' \gamma (\beta' + \beta) \\ \Lambda^1_0(\vec{\beta}'') = \Lambda^1_\lambda(\vec{\beta}) \Lambda^\lambda_0(\vec{\beta}') \\ \phantom{\Lambda^1_0(\vec{\beta}'')} = \Lambda^1_0(\beta) \Lambda^0_0(\beta') + \Lambda^1_1(\beta) \Lambda^1_0(\beta') + 0 + 0 \\ \phantom{\Lambda^1_0(\vec{\beta}'')} = -\gamma \gamma' (\beta + \beta') \end{array} \right. \implies \Lambda^1_0(\vec{\beta}'') = \Lambda^1_0(\vec{\beta}'') . \quad (3.52)
\end{aligned}$$

Similarly, the other elements of $\Lambda^\mu_\nu(\vec{\beta}'')$ and $\Lambda^\mu_\nu(\vec{\beta}'')$ also match. Thus Eq. (3.46) does not hold in this case.

Case II: $\vec{\beta} \perp \vec{\beta}'$

Let us choose $\vec{\beta}$ and $\vec{\beta}'$ to be directed along the positive x^1 axis and the positive x^2 respectively, i.e.

$$\vec{\beta} = (\beta, 0, 0) \quad \text{and} \quad \vec{\beta}' = (0, \beta', 0), \quad (3.53)$$

whence the non-vanishing elements of $\Lambda^\mu_\nu(\vec{\beta})$ would be the same as in Eq. (3.49), whereas the non-vanishing elements of $\Lambda^\mu_\nu(\vec{\beta}')$ are given by

$$\Lambda^\mu_\nu(\vec{\beta}') : \quad \begin{cases} \Lambda^0_0(\beta') = \Lambda^2_2(\beta') = \gamma' , \\ \Lambda^2_0(\beta') = \Lambda^0_2(\beta') = -\beta' \gamma' , \\ \Lambda^1_1(\beta') = \Lambda^3_3(\beta') = 1 , \end{cases} \quad \left[\gamma' = \frac{1}{\sqrt{1 - \beta'^2}} \right]. \quad (3.54)$$

Using Eqs. (3.45) and (3.47) the non-vanishing elements of $\Lambda^\mu_\nu(\vec{\beta}'')$ and $\Lambda^\mu_\nu(\vec{\beta}'')$ are now computed as:

$$\begin{aligned}
\text{0-0 element: } \left\{ \begin{array}{l} \Lambda^0_0(\vec{\beta}'') = \Lambda^0_\lambda(\vec{\beta}') \Lambda^\lambda_0(\vec{\beta}) \\ \phantom{\Lambda^0_0(\vec{\beta}'')} = \Lambda^0_0(\beta') \Lambda^0_0(\beta) + 0 + 0 + 0 = \gamma' \gamma \\ \Lambda^0_0(\vec{\beta}'') = \Lambda^0_\lambda(\vec{\beta}) \Lambda^\lambda_0(\vec{\beta}') \\ \phantom{\Lambda^0_0(\vec{\beta}'')} = \Lambda^0_0(\beta) \Lambda^0_0(\beta') + 0 + 0 + 0 = \gamma \gamma' \end{array} \right. \implies \Lambda^0_0(\vec{\beta}'') = \Lambda^0_0(\vec{\beta}'') . \quad (3.55)
\end{aligned}$$

But

$$\begin{aligned}
\text{1-0 element: } \left\{ \begin{array}{l} \Lambda^1_0(\vec{\beta}'') = \Lambda^1_\lambda(\vec{\beta}') \Lambda^\lambda_0(\vec{\beta}) \\ \phantom{\Lambda^1_0(\vec{\beta}'')} = 0 + \Lambda^1_1(\beta') \Lambda^1_0(\beta) + 0 + 0 = -\beta \gamma \\ \Lambda^1_0(\vec{\beta}'') = \Lambda^1_\lambda(\vec{\beta}) \Lambda^\lambda_0(\vec{\beta}') \\ \phantom{\Lambda^1_0(\vec{\beta}'')} = \Lambda^1_0(\beta) \Lambda^0_0(\beta') + 0 + 0 + 0 = -\beta \gamma \gamma' \end{array} \right. \implies \Lambda^1_0(\vec{\beta}'') \neq \Lambda^1_0(\vec{\beta}'') . \quad (3.56)
\end{aligned}$$

Similarly, a few other elements, such as 2-0, 1-2, ..., do not match as well, and hence verify Eq. (3.46).

3.7.2 Relativistic Velocity Addition

Refer back to Fig. 6, and the scenario it depicts. If the relative frame velocity $\vec{v}$ and the particle's velocity $\vec{v}'$ w.r.to the frame Σ' are known, then what would be the particle's velocity $\vec{v}''$ w.r.to the frame Σ ?

To determine this, let us consider the simple case in which $\vec{v}$ and $\vec{v}'$ are parallel. If we choose to take $\vec{v} = c\vec{\beta}$ and $\vec{v}' = c\vec{\beta}'$ to be directed along x^1 , then this is the case I considered in §3.7.1. From Fig. 6 it is obvious that $\vec{v} \parallel \vec{v}' \parallel x^1$ implies $\vec{v}''$ is also directed along x^1 . The problem thus becomes an (1+1)-dimensional one, rather than (3+1)-dimensional, as the $(x^0 - x^1)$ plane is of relevance only.

The non-vanishing elements of the transformation matrices $\Lambda^\mu_\nu(\vec{\beta})$ and $\Lambda^\mu_\nu(\vec{\beta}')$ are given in Eqs. (3.49) and (3.50) respectively. Since $\vec{\beta}'' = \vec{v}''/c = (\beta'', 0, 0)$, the non-zero elements of the transformation matrix

$\Lambda^\mu_\nu(\vec{\beta}'')$ are similar:

$$\Lambda^\mu_\nu(\vec{\beta}'') : \quad \begin{cases} \Lambda^0_0(\beta'') = \Lambda^1_1(\beta'') = \gamma'' , \\ \Lambda^1_0(\beta'') = \Lambda^0_1(\beta'') = -\beta'' \gamma'' , \\ \Lambda^2_2(\beta'') = \Lambda^3_3(\beta'') = 1 , \end{cases} \quad \left[\gamma'' = \frac{1}{\sqrt{1 - \beta''^2}} \right] . \quad (3.57)$$

Comparing the $\Lambda^0_0(\beta'')$ and $\Lambda^1_0(\beta'')$ with those in Eqs. (3.51) and (3.52) [case I of §3.7.1], we have

$$\gamma'' = \gamma' \gamma (\beta' + \beta) , \quad \beta'' \gamma' = \gamma' \gamma (\beta' + \beta) . \quad (3.58)$$

Solving simultaneously gives the *velocity addition formula* in this simple case in which the relative velocity $\vec{v}$ of the two frames Σ and Σ' is parallel to the particle's velocity $\vec{v}'$ w.r.to Σ' :

$$\boxed{\beta'' = \frac{\beta' + \beta}{1 + \beta' \beta}} \quad \Rightarrow \quad \boxed{v'' = \frac{v' + v}{1 + v'v/c^2}} . \quad (3.59)$$

For the general formula of velocity addition (when $\vec{v}$ is not parallel to $\vec{v}'$), see the derivation in Appendix B.